

H5N1 avian influenza genomic surveillance in domestic cats in the Americas

A.I. Maksiaev, A. Crespo-Bellido and M.I. Nelson

Virus Evolution Lab, National Library of Medicine, National Institutes of Health

Abstract

Since late 2021, clade 2.3.4.4b of the highly pathogenic avian influenza (HPAI) H5N1 virus has rapidly spread among various bird and mammal species across North America with devastating consequences for livestock and wildlife (1). In this time, the US Department of Agriculture has sequenced thousands of H5N1 genomes. However, not all H5N1 sequence data is deposited in one central location, creating barriers to data analysis. To address this problem, we built a genomic data pipeline to combine data from three influenza databases. This pipeline involves web-scraping (parsing both HTML and XML sites), de-duplicating between all three databases using isolates, and cleaning metadata and sequence data every week for rapid real-time phylogenetic analyses. Phylogenetic trees were inferred for emerging genotypes impacting livestock and humans (e.g., B3.13, D1.1). As a case study, we examined in detail the transmission of the B3.13 genotype to domestic cats in the United States. We identified 181 H5N1 genomes sequenced from cats across 24 US states. Through phylogenetic analysis we identified potential transmission sources. Going forward, this pipeline will be a useful tool for rapidly identifying interspecies transmission events, adaptive mutations that help H5N1 transmit in mammals, including humans, and potential interventions to mitigate risk.

Methods

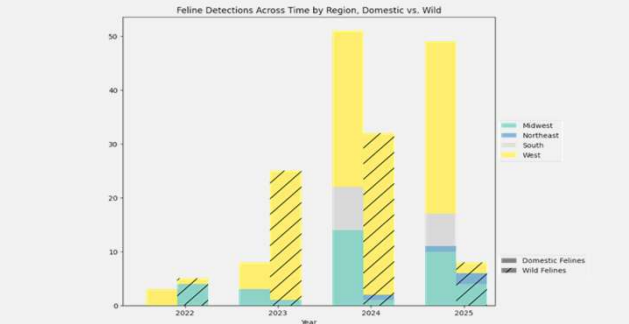
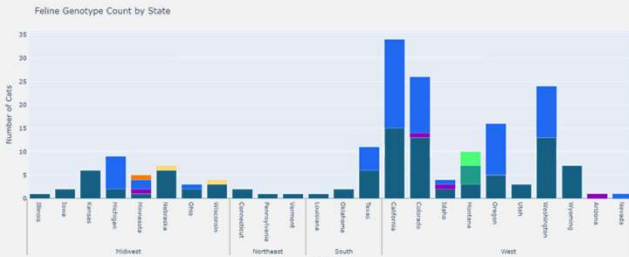
Creating phylogenetic trees:

Web-scraping: We downloaded all influenza A H5N1 genome sequences collected since Nov. 1, 2021, to Jul. 25, 2025, from the Western hemisphere (330 A3 sequences, 1896 B3.2 sequences, 955 B3.6 sequences, 122 B3.7 sequences, 4924 B3.13 sequences, and 3143 D1.1 sequences).
De-duplication: we used unique isolates and accessions to de-duplicate data such that if one unique sequenced genome appeared in multiple databases, only one was used for analysis. First, sequences were obtained from SRA, then de-duplicated and merged with sequences from GenBank, and then this merged dataset was de-duplicated and merged with sequences from GISAID (Figure 1).
Cleaning metadata: we created labels containing metadata and labeled all sequences collected.
Alignment: we used MAFFT via BioWulf and aligned sequences with like genotypes, separated by segment (6).
Trimming and concatenation: we trimmed the ends of each aligned sequence segment to only the coding regions, then concatenated the segments together.
Phylogenetic analysis: using IQ-TREE2 via BioWulf, we created a maximum-likelihood phylogenetic tree with GTR+G and ultrafast bootstrap with 1000 iterations per genotype (7).

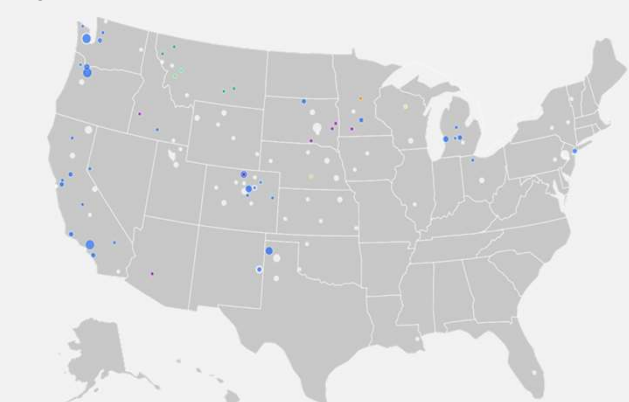
Results

Using the pipeline, we followed in detail the spread of various H5N1 genotypes to **wild and domestic cats** in the United States. We then identified 181 H5N1 genomes sequenced from wild and domestic cats and used geospatial USDA data to identify cases across 24 US states (Figure 2). Overall, detections seem to be rising over time, especially in the western region (Figure 3). This is also where the B3.13 genotype is most plentiful among felines (Figure 4).

Through phylogenetic analysis, we identified potential transmission sources with a focus on genotype B3.13, including contaminated raw milk from B3.13-infected **dairy cows**, **raw pet food** from B3.13-infected poultry, and caught B3.13-infected **wild birds** (Figure 5).



USDA Feline Avian Influenza Detections



References

- Xie Z, Yang J, Zhao W, Li X, Iqbal M, Liao M, Dai M. 2025. Clade 2.3.4.4b highly pathogenic avian influenza H5N1 viruses: knowns, unknowns, and challenges. *J Virol* 99:e00424-25. <https://doi.org/10.1128/jvi.00424-25>
- Detections of highly pathogenic avian influenza in mammals. Animal and Plant Health Inspection Service. September 3, 2025. Accessed September 3, 2025. <https://www.aphis.usda.gov/livestock-poultry-disease/avian/avian-influenza/avian-influenza-detections/mammals>
- NCBI virus. National Center for Biotechnology Information. Accessed September 3, 2025. https://www.ncbi.nlm.nih.gov/labs/virus/vssi/#/virus?SeqType_s=Nucleotide&VirusLineage_s=H5N1&Serotype_s=H5N1

Acknowledgements

This research was supported in part by the Intramural Research Program of the National Institutes of Health (NIH). We gratefully acknowledge all data contributors, i.e., the Authors and their Originating laboratories responsible for obtaining the specimens, and their Submitting laboratories for generating the genetic sequence and metadata and sharing via the GISAID Initiative, on which this research is based.

Introduction

Since first detecting the spread, The US Department of Agriculture (USDA) has sequenced thousands of clade 2.3.4.4b of the highly pathogenic avian influenza (HPAI) H5N1 virus genomes to track evolution, transmission, and risk to human and animal populations (2). However, not all H5N1 sequence data is deposited in **NCBI's GenBank**, creating barriers to data analysis (3). To address this problem, we built a genomic data pipeline to combine data from three influenza databases: **Sequence Read Archive (SRA)**, **GenBank**, and **GISAID** (4, 5).

Data sourced to create figures:

USDA mammal avian influenza detections: This dataset allowed us to obtain geospatial data and counts of reported feline cases of H5N1 throughout the US. Combining this dataset with the dataset used to create the phylogenetic trees (created from GenBank, SRA, and GISAID), we matched metadata (US state, collection date, and species) to estimate which counties cases of various genotypes were located in.

Figure 1. Workflow diagram of genomic data pipeline using SRA, GenBank, and GISAID.

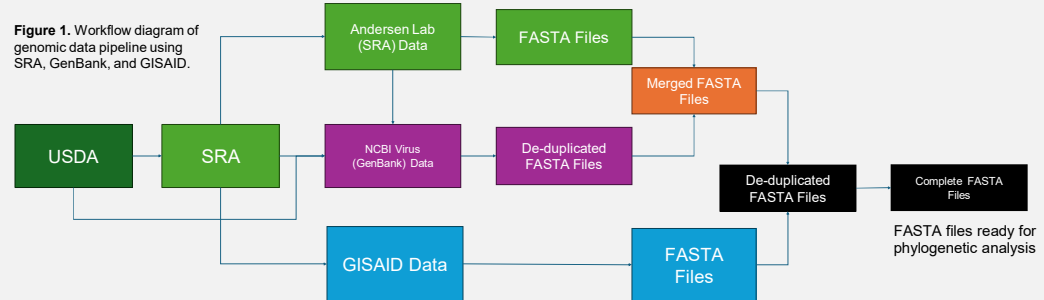


Figure 2. Bar plot showing USDA-reported virus detection count in felines by genotype per state and region between Nov. 1 2021 – Jul. 25 2025.



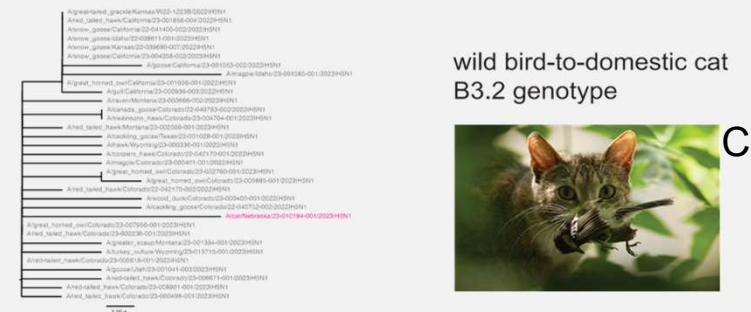
dairy cattle-to-domestic cat B3.13 genotype

Figure 3. Bar plot showing USDA-reported virus detection count in felines (domestic and wild) across time in years Nov. 1 2021 – Jul. 25 2025 by United States region.



raw pet food-to-domestic cat B3.13 genotype

Figure 4. United States map showing various genotypes of USDA-reported feline H5N1 cases detected between Nov. 1 2021 – Jul. 25 2025.



wild bird-to-domestic cat B3.2 genotype

Figure 5. Phylogenetic subtrees showing virus transmission between cattle and domestic cat (A), frozen pet food and domestic cat (B), and wild bird and domestic cat (C).

Conclusion

Through this pipeline, we have seen that avian influenza of the B3.13 genotype is frequently spilling over into domestic cats, particularly in Western regions where the virus circulates in dairy cattle. Through phylogenetic analysis, we have also seen the various transmission events through which **domestic cats** get infected: **raw milk**, **raw pet food**, and **wild birds**. However, this pipeline will be a useful tool for identifying potential interventions to mitigate risk, by rapidly identifying interspecies transmission events, and adaptive mutations that help H5N1 transmit in mammals, including humans.

- Avian-influenza: Consensus sequences for US H5N1 clade 2.3.4.4b. GitHub. 2025. Accessed September 3, 2025. <https://github.com/andersen-lab/avian-influenza>
- Elbe, S. and Buckland-Merrett, G. (2017) Data, disease and diplomacy: GISAID's innovative contribution to global health. *Global Challenges*, 1:33-46. doi:10.1002/gch2.1018
- Katoh K, Standley DM. MAFFT multiple sequence alignment software version 7: improvements in performance and usability. *Mol Biol Evol*. 2013;30(4):772-780. doi:10.1093/molbev/mst010.
- B.Q. Minh, H.A. Schmidt, D. Chernomor, D. Schrempf, M.D. Woodhams, A. von Haeseler, R. Lanfear (2020) IQ-TREE 2: New models and efficient methods for phylogenetic inference in the genomic era. *Mol. Biol. Evol.* 37:1530-1534.

Disclaimer

The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health or the U.S. Department of Health and Human Services, or the United States government.